SENTINEL AI: DETECTION ALGORITHMS BLUEPRINT

Deterministic Multi-Modal Sensor Fusion & Veto Architecture Specification

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1. EXECUTIVE SUMMARY & PROBLEM CONTEXT

Traditional emergency response systems (such as the default Android 5-power-button SOS or standalone acoustic scream monitors) suffer from catastrophic false-positive rates exceeding 90%. Overwhelming volumes of accidental pocket dials (caused by key compression inside bags or tight denim) and false alarms triggered by entertainment audio (screams in television movies or video games) frequently overwhelm police dispatchers.

The Sentinel AI 'Cross-Modal SOS Veto Engine' introduces an intelligent, deterministic software gate. Rather than continuously draining battery life by sampling 24/7, the system remains dormant until an interrupt occurs (such as 5 rapid power-button presses or an amplitude spike). It then captures a synchronized 2,000-millisecond multi-sensor snapshot across three physical modalities, applies deterministic signal processing equations, and evaluates whether to ASSERT VETO (cancel dispatch) or CONFIRM DISPATCH (connect directly to Helpline 1091).

2. THE THREE SENSOR MODALITIES (PHYSICAL REASONING)

To guarantee defensibility under real-world physical constraints and patent scrutiny, the engine relies on three distinct physical sensor domains:

Modality 1: Kinetic Inertial Motion (Accelerometer + Gyroscope)

- Measures physical force, impact dynamics, and spatial rotational variance.
- Distinguishes smooth walking rhythm or table rest from violent physical struggle.
- Key Metric: Vector Jerk J = da/dt in g/s and angular variance sigma_omega^2 in rad/s^2.

Modality 2: Optical Exposure Index (Proximity Sensor + Ambient Photometer)

- Identifies containment context: open atmosphere vs. trapped within dark enclosure.
- Evaluates physical proximity distance (binary covered vs. far) and ambient lux levels.
- Metric: O_containment = True when proximity is blocked and illuminance < 5.0 lux.

Modality 3: Acoustic Playback & Harmonic Filter (16-bit PCM @ 16 kHz)

- Analyzes spectral harmonic dispersion to eliminate television and radio playback.
- Leverages lossy compression rolloff: MP3/AAC codecs cut audio abruptly <= 15.0 kHz.
- Live biological vocal tracts produce wideband distress harmonics extending > 16.0 kHz.

3. CONCRETE MATHEMATICAL FORMULATIONS

Every decision is mathematically computed in real time inside the 2,000ms capture window without remote cloud calls:

A. Kinetic State Metric (K) & 3D Jerk Calculus

Static acceleration magnitude (mean_g) cannot distinguish aggressive physical assault from someone jumping or running down stairs. Sentinel AI computes the Jerk Vector (J), representing the first time-derivative of 3-axis acceleration:

Formula: J(t) = d(a(t))/dt = lim [a(t + dt) - a(t)] / dt

Euclidean 3-Axis Jerk Magnitude: |J| = sqrt((dAx/dt)^2 + (dAy/dt)^2 + (dAz/dt)^2)

Rotational Variance across N Gyroscope samples: sigma_omega^2 = (1/N) * sum((||omega_i|| - mean_omega)^2)

- Resting Table Condition: |J| < 0.20 g/s and mean_g ~ 1.0 g.
- Walking Rhythm: 0.20 g/s <= |J| < 2.0 g/s and mean_g < 1.5 g.
- Violent Physical Struggle / Blows: |J| > 3.0 g/s and rotational variance > 3.5 rad/s^2.

B. Optical Containment Index (O)

Quantifies probability that the mobile device is trapped inside clothing or bag fabric:

O_index = 1.0 if (Prox == Covered and Lux < 5.0) else (0.0 if Lux >= 20.0 else (20 - Lux)/20)

C. Acoustic Spectral Rolloff & Total Harmonic Distortion (THD)

The audio engine executes FFT spectrum analysis over 16-bit PCM streaming buffers to inspect harmonic decay:

THD Equation: THD(%) = (sqrt(V2^2 + V3^2 + V4^2 + ...) / V1) * 100

- Codec Low-Pass Rolloff: Digital television, streaming video, and loudspeaker transducers exhibit steep low-pass filtering with cutoff frequencies fc <= 15.0 kHz.
- Biological Human Distress Scream: Authentic human screams produce high-energy acoustic turbulence across the vocal folds with spectral bandwidth extending well beyond 16.0 kHz up to 20.0 kHz.

D. Cross-Modal Confidence Score (CMCS)

When telemetry inputs fall into intermediate or ambiguous zones, the engine applies weighted fusion:

CMCS = (w_k * K_score) + (w_o * O_score) + (w_a * A_score)

Where: w_k = 0.45 (Kinetic), w_o = 0.25 (Optical), w_a = 0.30 (Acoustic). Threshold CMCS >= 0.40 dispatches emergency; < 0.40 asserts veto.

4. COMPREHENSIVE SENSOR TELEMETRY COLUMNS SCHEMA

The system exposes deterministic column values across every evaluation snapshot, stored in Room SQLite:

Domain	Column Name	Data Type	Nominal Range	Physical Significance
Kinetic	mean_g	Float (g)	0.8 - 3.5 g	Average gravitational acceleration vector
Kinetic	jerk	Float (g/s)	0.0 - 8.0 g/s	Rate of change of force; struggle intensity
Kinetic	rotational_var	Float (rad/s^2)	0.0 - 10.0	Gyro variance; erratic fighting tumbling
Kinetic	motion_type	String	Enum	STATIONARY, RHYTHMIC_WALKING, VIOLENT_STRUGG
Optical	prox_covered	Boolean	True / False	Infrared proximity obstruction flag (pocket)
Optical	ambient_lux	Float (lx)	0.0 - 10k lx	Ambient photometer (open air vs enclosed)
Acoustic	thd_percentage	Float (%)	1.0 - 50.0%	Total Harmonic Distortion (speaker resonance)
Acoustic	high_freq_cutoff	Float (kHz)	8.0 - 20.0 kHz	Spectral rolloff (TV codec vs bio scream)
Acoustic	voice_stress	Boolean	True / False	Vocal fold distress classifier
Engine	action	String	VETO/DISPATCH	Deterministic arbitration decision outcome

5. REAL-WORLD SCENARIOS & DECISION RULES

Scenario A: Pocket / Bag Accidental Dial (ASSERT VETO)

? Physical Trigger: Power button pressed 5 times while walking or phone bouncing inside handbag.

? Telemetry State: Optical proximity covered (True), Ambient illuminance = 0.05 lx, Jerk = 0.65 g/s (< 2.0), Voice stress = False.

? Decision Output: ACTION = 'VETO', Reason = 'ACCIDENTAL_POCKET_DIAL', Confidence = 97%, Haptic = 'SILENT_ABORT'.

? Result: Baseband 1091 emergency call suppressed without disturbing police control desks.

Scenario B: Television / Entertainment Scream (ASSERT VETO)

? Physical Trigger: Acoustic spike detector triggered by a movie, gaming stream, or video playback.

? Telemetry State: Kinetic IMU indicates stationary rest on table (Jerk = 0.08 g/s, mean_g = 1.01), Audio spectral cutoff = 14.2 kHz (<= 15.0 kHz codec cutoff).

? Decision Output: ACTION = 'VETO', Reason = 'MEDIA_PLAYBACK_DETECTED', Confidence = 93%, Haptic = 'SILENT_ABORT'.

? Result: Loudspeaker scream rejected as artificial media; alarm silently aborted.

Scenario C1: Violent Physical Assault / Struggle (CONFIRM DISPATCH)

? Physical Trigger: Power button pressed or shake triggered while victim is actively attacked.

? Telemetry State: Kinetic Jerk = 4.82 g/s (> 3.0), Rotational variance = 6.15 rad/s^2, mean_g = 2.45 g.

? Decision Output: ACTION = 'DISPATCH', Confidence = 96%, Countdown = 0s, Haptic = 'DISTRESS_CONFIRMED'.

? Result: Zero-countdown emergency call to 1091, loud siren alarm, offline SMS dispatch, audio evidence recording.

Scenario C2: Authentic Human Distress Scream (CONFIRM DISPATCH)

? Physical Trigger: Woman vocalizes high-amplitude scream for help.

? Telemetry State: Voice stress = True, High frequency harmonic cutoff = 19.4 kHz (> 16.0 kHz uncompressed vocal folds).

? Decision Output: ACTION = 'DISPATCH', Confidence = 95%, Countdown = 0s, Haptic = 'DISTRESS_CONFIRMED'.

? Result: Immediate connection to Singappen Police Control Room.

6. PATENT CLAIM DRAFTING BLUEPRINT (35 U.S.C. Section 101)

To protect Sentinel AI's technical intellectual property, the method claims focus strictly on physical signal transformations rather than abstract machine learning:

> Independent System Claim 1: A mobile computing apparatus comprising: an inertial measurement unit, an optical sensor array, an acoustic transducer, a cellular baseband transceiver, and an execution processor coupled to memory, wherein the processor is configured to: detect an emergency trigger event while in a low-power dormant state; open a temporal micro-burst capture window of 2,000 milliseconds querying the inertial, optical, and acoustic transducers; compute an inertial jerk vector J = da/dt; compute an optical containment parameter; compute an acoustic frequency cutoff parameter; programmatically assert a veto command suppressing initiation of an outgoing baseband emergency dispatch to a remote police terminal upon determining the optical sensor is obstructed, the jerk vector remains below 2.0 g/s, and acoustic voice stress is absent; and selectively authorize baseband emergency dispatch upon detecting either jerk magnitude exceeding 3.0 g/s or an acoustic spectral rolloff exceeding 16.0 kHz.